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PREDICTION OF CRITICAL HEAT FLUX OF WIRE-WRAPPED ROD BUNDLE BASED ON ARTIFICIAL NEURAL NETWORK

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ABSTRACT

Accurate prediction for the critical heat flux of rod bundle assemblies is important for reactor safety assessment. However, the existing prediction methods fail to unify their generalization and accuracy. The theoretical prediction models are unable to provide a satisfactory estimation. while the empirical methods that have been proposed for several decades could give results precisely, their suitable conditions are strictly limited, causing the endless development of different correlations for every specific rod bundle design. An exploration of new methods has been undertaken to provide a better method for critical heat flux prediction based on an artificial neural network since neural networks have gained recent attention for their outstanding performance in predicting complex problems. This study discusses the prediction performance of a traditional polynomial fitting method, a neural network method, and a hybrid method combined with empirical correlations and a fully connected neural network regression model. The analysis concludes that when artificial neural networks replace empirical models in prediction tasks, achieving equivalent prediction accuracy is easily accomplished; however, there is a need for improvement in error characteristics. Besides, the hybrid method was validated as a superior approach, demonstrating its effectiveness in enhancing prediction performance across various input variable regions. The ratio of predicted values to true values is well-maintained, indicating that this proposed method yields improved error characteristics, and consequently, more accurate critical heat flux estimations. The research will help improve the feasibility of the application of machine learning algorithms such as neural networks in the nuclear energy industry.

Keywords: Critical heat flux, Artificial neural network, Error analysis

NOMENCLATURE

G	mass flux at the critical point
Δh	enthalpy difference between the critical point and the inlet of the test section
p	pressure at the critical point
q	critical heat flux
q_P	predicted value of critical heat flux
q_M	measured value of critical heat flux
x	input variable
x_{eq}	steam quality at the critical point
y	true value of the prediction task
y_{pred}	output variable of the hybrid method
$\hat{y}$	output variable of the prior model
ϵ	difference between y and $\hat{y}$
$\hat{\epsilon}$	output variable of the regression model

1. INTRODUCTION

The pursuit of the limit in heat transfer performance has made the prediction of critical heat transfer a continual focus in thermal-hydraulic research. This is crucial across various industries, including heat dissipation in electronics[1], material flammability[2], nuclear engineering[3], and other boiler applications[4]. In pressurized water reactors, the fuel rod bundle in the reactor core generates energy, and the coolant in the first loop continuously transfers the heat generated by the rod bundle downstream. The critical phenomenon occurs when a transition in boiling mechanisms at the interface between the heat source and the coolant prevents the heat transfer into the coolant. Due to the abrupt change in the heat transfer coefficient, once the critical state is reached, the deterioration of the heat transfer can rapidly elevate the temperature of the rod bundle, leading to severe consequences such as core meltdown. This significantly impacts the safe operation of the reactor. Therefore, accurately predicting critical heat transfer is crucial for the economic and safe operation of the reactor.

Over the years, researchers have conducted extensive studies, providing numerous models based on critical prediction mechanisms and empirical models[5-10]. Mechanistic models are developed based on an understanding of boiling heat transfer mechanisms. In a mechanistic model, a mathematical representation of the critical triggering mechanism is

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established, and suitable bubble dynamics models are employed to refine this critical mechanism model. However, the generalizability of existing mechanistic models is typically limited, and they are often not directly applicable to rod bundle critical prediction. In fact, our understanding of critical mechanisms remains limited[11, 12]. Various mechanistic models are only applicable within specific ranges. Predicting critical points for rod bundle structures often proves challenging and yields less accurate results. On the other hand, empirical models[13-15], are typically data-driven fitting formulas representing a set of well-measured experimental data. Therefore, empirical models, or empirical correlations, exhibit relatively high prediction accuracy but are restricted to the scope claimed, including thermal-hydraulic conditions, structures, and subchannel codes. The empirical model owns no generalization capability.

In recent years, with the flourishing development of deep learning research[16-18], the effectiveness of neural networks as regression models has been verified. However, different from previous approaches[19], relying on large models for regression may not be feasible due to data limitations. As an alternative, hybrid models, requiring less data and offering clearer physical significance, emerge as a potentially viable engineering application method. The hybrid method is the combination of the traditional model and the machine learning(ML) model. For heat transfer prediction, Suh et al.[20] provided a deep neural network framework to predict boiling heat transfer in the departure of nucleate boiling regime. The framework leverages cutting-edge deep learning models including convolutional neural networks and object detection algorithms to automatically extract both hierarchical and physics-based features, predicting the boiling curve with a mean error of 6%. Zhao et al.[21] suggested a hybrid framework for the prediction of departure from nucleate boiling. The hybrid framework takes advantage of established understanding in the field and uses machine learning models to capture the undiscovered information from the mismatch between experimental and domain knowledge-predicted output. It is claimed that the proposed hybrid approach has superior performance over conventional prediction models and standalone ML methods for both interpolation and extrapolation purposes. Also, Niu et al.[22] applied the deep feed-forward neural network to critical heat flux (CHF) prediction task in narrow channels. Compared with the traditional empirical method, the hybrid method can predict both the value of CHF and the specific location where the CHF occurs. The prediction error decreases from 10% to 5% compared to the standalone neural network model.

In this research, the hybrid method is applied to the prediction of the critical heat flux of a wire-wrapped rod bundle. The framework of the hybrid method consists of a prior model and an ML regression model. Empirical correlations are used as the prior model and the fully connected neural network is the regression model. Discussions are conducted based on the estimation performance of hybrid models. The methods will be helpful for the further development of CHF prediction in rod bundles.

2. MATERIALS AND METHODS

2.1 The hybrid method

As mentioned earlier, the hybrid method is an approach that involves a prior model with an existing understanding of critical heat transfer and an ML regression model with efficient approximation capabilities. Considering the framework in literature[21, 22], we present a framework for the hybrid method, as illustrated in Figure 1. The input, x, serves as input for both the prior model and the regression model. The input, y, is employed as an additional input during the training process of the regression model, as indicated by the green line in Figure 1. The prior model takes x as input and produces $\hat{y}$ as a baseline. The regression model, taking x as input, produces $\hat{\epsilon}$ as a correction. The final output, y_{pred}, is obtained by multiplying these two output values. In this method, the prior model provides a foundational solution to the problem, while the role of the regression model is closer to that of a correction factor.

In this paper, the CHF data comes from the experiment conducted by SJTU-NETH[23]. The test section is a 19-rod wire-wrapped rod bundle. Hence, our prior model employs the W-3 correlation[13] and KfK-3 correlation[24]. For uniform input, x is represented as local pressure p, local mass flux G, steam quality x_{eq}, enthalpy difference Δh, and the output value y_{pred} corresponds to critical heat flux q. The predictive performance of the model will be detailed in Section 3.

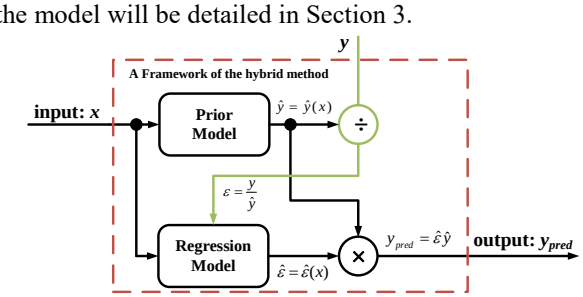

FIGURE 1: A FRAMEWORK OF THE HYBRID METHOD

2.2 Evaluation of the hybrid method

The model evaluation is manifested through the difference between predicted values (q_P) and true values (q_M). In this study, the distribution is plotted with predicted values on the x-axis and true values on the y-axis to visualize the predictive performance. Ideally, all points are expected to be distributed near the diagonal, and the maximum percentage deviation boundaries represent the predictive capability lower limits of the model. Furthermore, error characteristics are discussed through the ratio of q_P and q_M. Typically, q_M/q_P is expected to follow a normal distribution and remain unaffected by changes in input variables. This indicates that the errors stem from random variables rather than amplifying under different input conditions. The distribution of error (q_M/q_P-1) will also be discussed to explicitly delineate the error characteristics of different methods.

3. RESULTS AND DISCUSSION

This section presents the predictive capabilities of the hybrid method together with discussions. Firstly, we provide the

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predictive outcomes of conventional empirical correlations. Considering the data used, the KfK-3 correlation[24, 25], which is applicable to those with a wire-wrapped spacer, is selected as the basis.

As shown in Figure 2, the KfK-3 correlation exhibits an overall underestimation when predicting the CHF of our experimental conditions. This is because the geometric structure for which the KfK-3 correlation is applicable differs from the tested. therefore, upon incorporating our geometric parameters in the correlation, there exists a slight deviation in predictions.

Also, the W-3 correlation is a commonly used empirical model for predicting critical heat flux and is applicable to both round tubes and rod bundles. After incorporating our geometric parameters into the W-3 correlation, its predicted values are significantly higher than the experimental values, resulting in an excessive overestimation of the experimental data for wire-wrapped rod bundles.

Hence, it is evident that for any specific rod bundle structure, the development of a predictive correlation tailored to the corresponding rod bundle structure is essential to achieve accurate predictions.

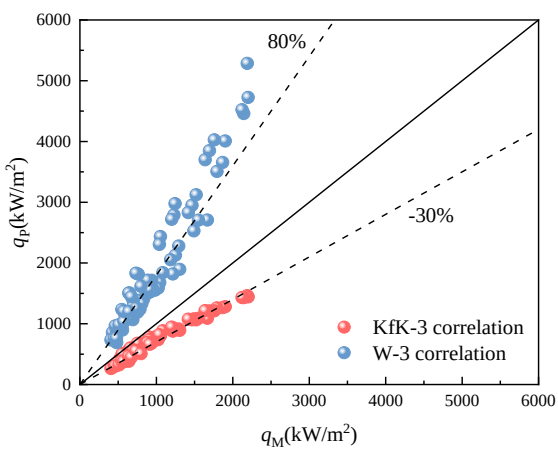
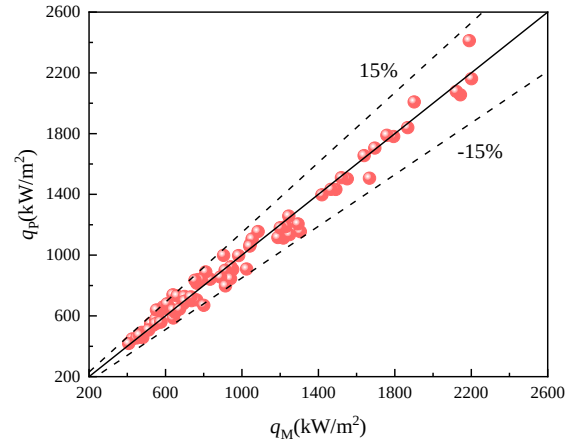


FIGURE 2: PERFORMANCE OF EMPIRICAL CORRELATIONS

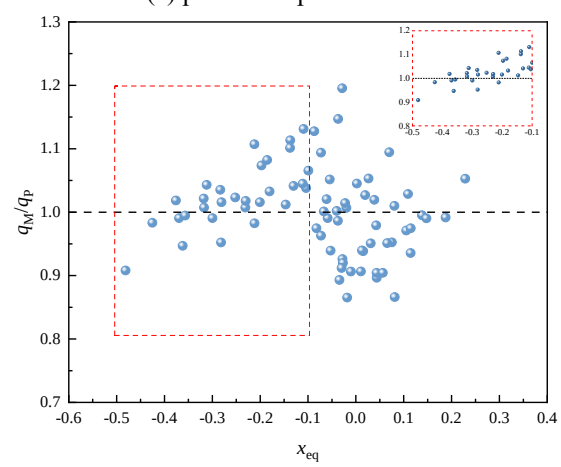
Next, we employ a polynomial fitting method to develop a suitable empirical model. Typically, when developing an empirical model for critical heat flux(CHF), thorough consideration of the polynomial form is necessary to avoid limitations on predictive capabilities imposed by the nature of its terms. To streamline the process, the formula structure is based on the FC correlation[26], with the exclusion of identification parameters used to describe grid types. The formula is expressed as follows, where a_i and b_i represent the parameters to be optimized:

$$q_{CHF} = a_1 + a_2 x_{eq} + a_3 G + a_4 (P^2 + b_1 P + b_2)^{b_3} + a_5 x_{eq} G + a_6 PG \quad (1)$$

The predictive results are illustrated in Figure 3. The empirical model exhibits relatively good predictive performance, with its prediction superior to directly applying other correlations. The error characteristics, represented by the ratio of q_M and q_P, are relatively satisfactory, generally within the range of 0.8 to 1.2. As shown in Figure 3(b), in the lower steam quality range, there is a linear increase in error as humidity rises. This suggests that as the polynomial fitting method is further extended, concerns may arise regarding the potential for significant predictive errors.



(a) prediction performance



(b) error characteristics as a function of variable x_{eq}

FIGURE 3: PERFORMANCE OF POLYNOMIAL FITTING CORRELATION

The results of the standalone neural network are illustrated in Figure 4. The neural network structure employed is 4-10-10-5-1, comprising a total of 25 neurons distributed across 3 layers with 10, 10, and 5 neurons in each layer, respectively. The predictive performance of the neural network is superior compared to traditional polynomial fitting. The predicted mean is closer to 1, and the variance is smaller. While its error characteristics are relatively better, the prediction errors are still influenced by pressure, as depicted in Figure 4(b).

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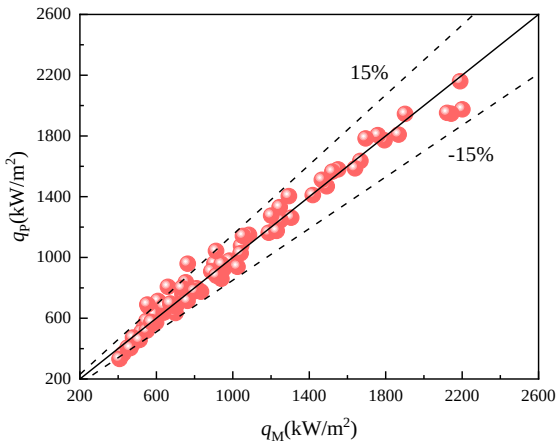
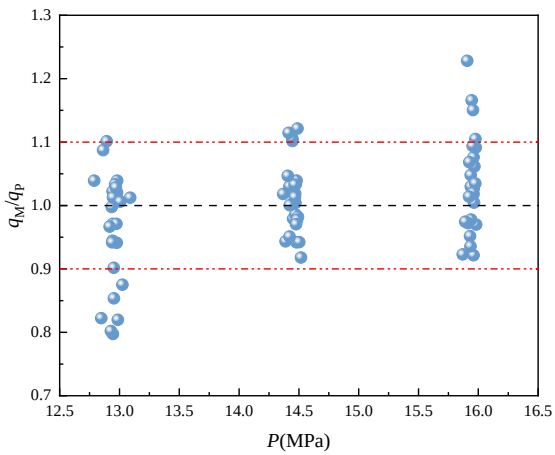


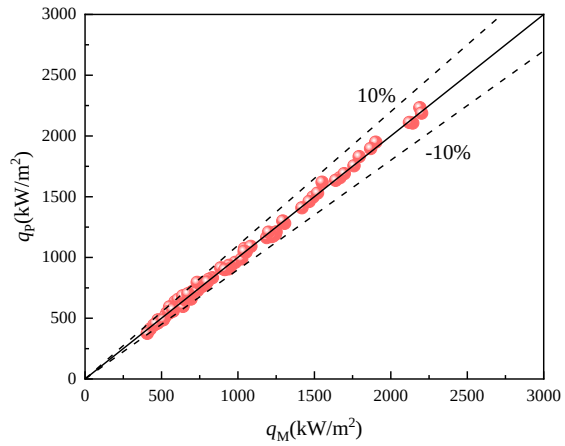
(a) prediction performance



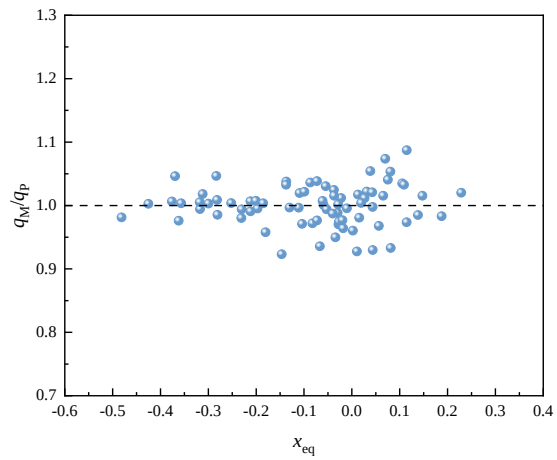
(b) error characteristics as a function of variable P

FIGURE 4: PERFORMANCE OF STANDALONE NEURAL NETWORK

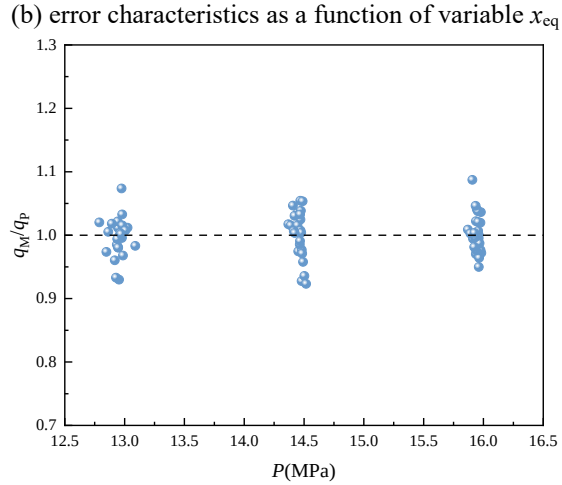
The results of the hybrid method are illustrated in Figure 5. The KfK-3 relationship is employed as the prior model and a fully connected neural network (the same architecture as before) as the regression model for correction coefficients. The predictive performance of the hybrid method, as shown in Figure 5, significantly improves through the combination of the two aforementioned models. The error characteristics also exhibit improved results as shown in Figure 5(b)&(c). The hybrid method is more efficient compared to traditional polynomial fitting approaches, as the Universal Approximation Theorem[27] eliminates the need for a detailed discussion of formulaic forms. Moreover, the hybrid method is more accurate than a standalone neural network because it obtains the baseline value for the CHF problem through the prior model, leaving complex and difficult-to-identify information to the neural network.



(a) prediction performance



(b) error characteristics as a function of variable x_{eq}



(c) error characteristics as a function of variable P

FIGURE 5: PERFORMANCE OF THE HYBRID METHOD

Finally, the error distribution among different methods is compared. Figure 6 illustrates the fractions of errors of various magnitudes which is $y = \text{fraction}(\left|\frac{q_M}{q_P} - 1\right| < x)$. It is evident that the results of polynomial fitting and neural network methods are similar, with their error fractions being comparable to

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varying degrees, indicating similar levels of predictive capability. In contrast, the hybrid method exhibits the best error characteristics, outperforming the aforementioned two prediction methods, demonstrating the superiority of the hybrid method in prediction.

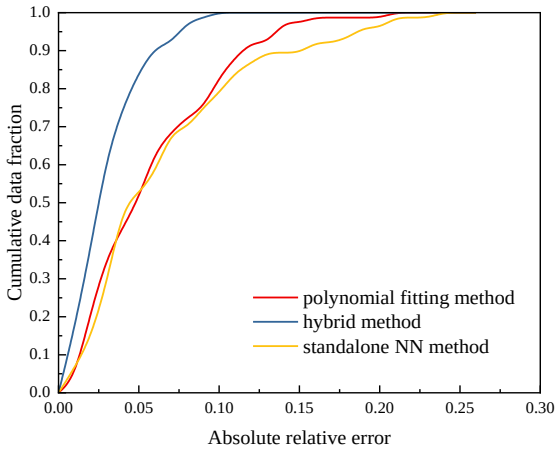

FIGURE 6: ERROR CHARACTERISTICS OF THE TESTED METHODS

4. CONCLUSION

In this paper, the predictive capabilities of the hybrid method in predicting critical heat flux for wire-wrapped rod bundles are discussed together with the advantages of the hybrid method. The hybrid method proves to be more efficient than traditional polynomial approaches, as structural optimization discussions are not required. Compared to the standalone neural network method, the hybrid method, incorporating a prior model, leverages prior knowledge, resulting in higher precision predictions at the same scale. Furthermore, the hybrid method outperforms the polynomial methods and standalone neural network in terms of both error characteristics and prediction accuracy.

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